MILITARY SPECIFICATION

FUZE, ROCKET: M439
PARTS AND ASSEMBLY WITH DETONATOR

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements and provides quality assurance provisions for the fabrication of parts and the manufacture, assembly and packing of fuze rocket: M439, for the 2.75 Inch Folding Fin Aerial Rocket.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-D-45495 - Detonator, Electric M84, Parts for and Loading, Assembling and Packing
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

FSC 1340

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-202 - Test Methods, Electrical and Electronic Components
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1167 - Ammunition Data Card

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402.)

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT AND PACKAGE DRAWINGS

9230116 - Box, Packing - Ammunition for Rocket 2.75 Inch
9235840 - Box, Cleated Plywood, Packing Ammunition for Rocket 2.75 Inch (Warhead M229, XM243, XM245 or XM255)
9260704 - Fuze, Rocket, M439
9270749 - Box, Fiberboard (Inner) Packing, Ammunition, for Fuze Rocket M439
9270750 - Box, Fiberboard (Outer) Alternative Packing, Ammunition, for Fuze Rocket M439
9270751 - Box, Packing, Ammunition for Fuze Rocket M439

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article inspection. This specification provides for a first article inspection. Requirements for the submission of the first article sample by the contractor shall be as specified in the contract.

3.1.1 Process specification plastic parts. When a multiple cavity mold is used in the fabrication of plastic components, five (5) components from each mold cavity shall be included in the First Article Sample. The components shall bear a mark which identifies the component with the individual cavities.

3.2 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.3 Molded or plastic components. All molded and plastic components shall be in accordance with applicable drawings and specifications.

3.4 Fuze assemblies and parts. The fuze assemblies and parts shall comply with all requirements specified on Drawing 9260704 and associated drawings and with all requirements specified in applicable standards and specifications.

3.5 Spring embrittlement. None of the springs used in the fabrication of the Safety and Arming Device (S&A) shall break when extended to the point of minimum permament distortion.

3.6 Spring set. The setback spring after being subjected to a temperature of 175 - 180°F at a compressed length of .750 inch for 24 hours and then allowed to cool at ambient for 24 hours, shall return to within .015 inch of its original free length.

3.7 Setback weight return capability. The setback weight in the S&A, after being bottomed to the base of the rotor housing by a force of 20 pounds applied equally to its two top surfaces and then released, shall return in a continuous motion to the fully safe position of locking the rotor.

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3.8 Rotor reset capability. The rotor assembly in the S&A, after being rotated between 26 degrees and 35 degrees from the safe position and then released, shall without external assistance and in a continuous motion be returned by the spring loaded setback weight to its fully safe and locked position.

3.9 Height of rotor assembly. The height of rotor above the inside surface of the rotor housing, with rotor in safe position, shall be 1.030 inches maximum.

3.10 Separation of annular gear from number 2 gear & pinion assembly. The separation distance between the annular gear and number 2 gear & pinion, when the rotor and number 2 gear are at their extreme separated position in the S&A, shall be .038 inch maximum.

3.11 Alignment of detonator cavity. The detonator cavity when the rotor is in the free armed detented position shall be in line with the outside diameter of the rotor housing within .060 in Total Indicator Reading (TIR) functionally.

3.12 Continuity and insulation resistance. The S&A assembly, shall satisfy the continuity and insulation requirements specified in Table I of this specification.

TABLE I. S&A continuity and insulation requirements.

Position of S&A Assembly	Terminals	Requirements
Safe	#1 & #2 (Charge Switch)	30 Megohms Min. 500 VDC
Safe	#1 & #5 (Charge Switch)	100 Milliohms Max.
Armed	#1 & #5 (Charge Switch)	30 Megohms Min. 500 VDC
Armed	#1 & #2 (Charge Switch)	100 Milliohms Max.
Armed	#4 & #6 & S&A Housing	100 Milliohms Max.

3.13 Detonators. Detonators, prior to assembling to rotors, shall meet the requirements of MIL-D-45495.

3.14 Loaded S&A assembly. Appropriate safety precautions shall be taken in testing for the following requirements.

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3.14.1 Load test of detonator. The detonator shall not move more than .005 inch when a push out force of 5 pounds is applied to the detonator socket.

3.14.2 Continuity through detonator (armed). The resistance measured between terminals 3 and 4 shall be 3.5 ± 1.5 ohms.

3.14.3 Non arming. The rotor assembly in the loaded S&A shall not arm when subjected to an arming acceleration of 12.0 ± 0.5 g's minimum for a period of 5 seconds. This test may be performed with a simulated detonator.

3.14.4 S&A arming. The S&A assembly, after being temperature conditioned, shall meet the following requirements:

3.14.4.1 Ambient temperature arming. The S&A assembly, after conditioning at a temperature of 70° ± 15°F for 2 hours minimum, shall arm between 0.70 and 0.90 second when subjected to an arming acceleration of 55.0 ± 0.5 g.

3.14.4.2 Low temperature arming. The S&A assembly after being conditioned at a temperature of -50° ± 5°F for 3 hours minimum, shall arm between 0.70 and 0.90 second when subjected to an arming acceleration of 55.0 ± 0.5 g.

3.14.4.3 High temperature arming. The S&A assembly after being conditioned at a temperature of +160° ± 5°F for 3 hours minimum shall arm between 0.70 and 0.90 second when subjected to an arming acceleration of 55.0 ± 0.5 g.

3.15 Molded Electronics assembly (module).

3.15.1 Module turn-on time. With the firing circuit energized, an output pulse shall be produced within 600 microseconds after activation of the timer.

3.15.2 Module time delay function. The module shall function within ± 2.0 percent of the set time. The firing output pulse shall provide at least 1000 ergs in less than 100 microseconds into an M84 detonator or equivalent load.

3.16 Module thermal shock. The module shall withstand the thermal shock test per MIL-STD-331, Test 113.

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3.17 Umbilical assembly. The insulation resistance shall be greater than 10 megohms and the continuity shall be less than 0.5 ohm when tested in accordance with MIL-STD-202.

3.18 S&A position. The safe position of the S&A shall be verified just before insertion into the fuze. After insertion of the S&A, fuze handling shall be controlled to preclude any chance of inadvertent arming.

3.19 Charge circuit continuity (fuze unarmed). - The continuity of the charge circuit through its umbilical terminals shall be 220 ohms minimum and 3.3k ohms maximum.

3.20 Environmental conditioning. The fuze shall meet all performance requirements after being subject to the following environmental conditions.

3.20.1 Temperature and humidity. Temperature and humidity in accordance with MIL-STD-331, Test 105.1, limited to one 14 day cycle.

3.20.2 Thermal shock. Thermal Shock in accordance with MIL-STD-331, Test 113.

3.21 Safety.

3.21.1 Jolt and jumble. The fuze shall be subjected to and must withstand the Jolt Test specified in MIL-STD-331, with the exception that the fuze shall not be disassembled and inspected after the Jolt Test. After completion of the Jolt Test, the same fuze shall be subjected to and must withstand the Jumble Test specified in MIL-STD-331.

3.21.2 Forty-foot drop. The complete fuze, warhead and motor assembly shall withstand the Forty-Foot Drop Test specified in MIL-STD-331. To satisfy the requirements of MIL-STD-331, the disassembled fuze must be safe to handle (see 6.6.1).

3.21.3 Five-foot drop. The fuze shall be safe to handle, fire or dispose of following the Five-Foot Drop Test. It need not be operable.

3.21.4 Non-arming. The fuze shall not arm when subject to an arming acceleration of 12.0 $\pm$ 0.5 g for a minimum of 5 seconds.

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3.21.5 Mechanical interrupt. The fuze shall be incapable of being charged in the armed condition.

3.22 Fuze operational. The fuzes shall be subjected to and must withstand the following operational tests: Non-Environmental Test - Non-Arm and Arming Times only, Helicopter Vibration Test (See Figure 1), Ballistic Test.

3.23 Fuze arming. The fuze, after being temperature conditioned, shall meet the following requirements:

3.23.1 Ambient temperature arming time. The fuze assembly, after conditioning at a temperature of $70^{\circ} \pm 15^{\circ}F$ for 2 hours minimum, shall arm between 0.70 and 0.90 second when subjected to an arming acceleration of 55.0 ± 0.5 g.

3.23.2 Low temperature arming time. The fuze after being conditioned at a temperature of $-50^{\circ} \pm 5^{\circ}F$ for 8 hours minimum, shall arm between 0.70 and 0.90 second when subjected to an arming acceleration of 55.0 ± 0.5 g.

3.23.3 High temperature arming time. The fuze after being conditioned at a temperature of $+160^{\circ} \pm 5^{\circ}F$ for 8 hours minimum shall arm between 0.70 and 0.90 second when subjected to an arming acceleration of 55.0 ± 0.5 g.

3.24 Charge-arm-function. The fuze shall be charged through its umbilical terminals, armed in accordance with fuze requirement 3.23.1 and shall, upon the closure of its setback weight switch, exhibit corresponding function time delays as shown in Table VI, Paragraph 2.

3.25 Reliability. The fuze shall demonstrate a minimum acceptable value of 96 percent with a best operational capability of 98 percent at 90 percent confidence.

3.26 Ballistic firing. The fuze shall function within $\pm$ 2 percent or 125 milliseconds of the set time whichever is greater after motor ignition. When properly charged and armed, the fuze shall function the detonator high order on expiration of the preset time delay.

3.27 Data requirements. The contractor shall generate data in accordance with the requirements of the data item descriptions cited in 4.5.37.

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3.28 Workmanship.

3.28.1 Mechanical. All parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, porosity, warpage, burn marks, checks, blisters, excess flash, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Surface coating shall be continuous except for a few slight scratches not exposing base materials.

3.28.2 Electrical. Cable assemblies and wires shall be dressed in such manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection.

b. Quality Conformance Inspection.

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Spring, Detent	8883713	10
Spring, Setback Weight	8883718	20
Strap, Rotor Ground	9260798	20
Speed Clip	9260891	20
Stud, Spring	8883721	10
Rotor Housing	9260795	10
Socket, Detonator	9260799	25
Ground Contact, S&A	9352734	25
Contact, common	9260783	25
Contact, N.C.	9260785	25
Contact, Detonator	9260787	25
Contact, Ground	9260788	25
Contact, N.O.	9260784	25
Plate, Top	9260781	25 or 5 per cavity(whichever is greater)
Plate, Bottom	9260782	25 or 5 per cavity (whichever is greater)
Gear and Pinion Assembly No. 1	8883703	25
Gear and Pinion Assembly No. 2	8883700	25
Escapement Gear and Pinion Assembly	8883710	25
Lever Assembly	9260776	30
Rotor Assembly	9260796	20
Rotor Housing Assembly	9260794	10
Inner Plate Assembly	9260804	20
Escapement Assembly	9260774	20
Setback Weight Assembly	9260790	25
Bottom Plate Assembly	9260780	25
S/A Assembly w/o Switch	9260779	25
S/A Assembly (M439 Rocket Fuze)	9260778	45
S/A Housing	9260699	25

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Pin, Interfacing	9260697	40
Electronic Assembly	9260689	25
Molded Electronics Assembly (Mold)	9260700	25 or 5 per cavity (whichever is greater)
Molded Electronics Assembly (Module)	9260700	810
Umbilical Ground Pin	9260719	25
Umbilical Charge Pin	9260720	25
Electronics Housing	9260698	25 or 5 per cavity (whichever is greater)
Electronic Housing Assembly	9260721	25
Fuze Rocket, M439	9260704	246

4.3.2 Inspections to be performed. See MIL-A-48078 and tables II, V and VI specified herein.

4.3.3 Rejection. See MIL-A-48078 and 4.5.36.3 herein.

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TABLE II. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE: Fuze Rocket, M439		SHEET 1 OF 10		DRAWING NUMBER: Listed below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Spring, Detent (Dwg. 8883713)				
	Examination for Defects	10		3.4	4.4.2.1
	Spring Embrittlement	10		3.5	4.5.3
	Spring, Setback Weight (Dwg. 8883718)				
	Examination for Defects	20		3.4	4.4.2.2
	Spring Embrittlement	10		3.5	4.5.3
	Setback Weight Spring Set	10		3.6	4.5.4
	Strap, Rotor Ground (Dwg. 9260798)				
	Examination for Defects	20		3.4	4.4.2.3
	Speed Clip (Dwg. 9260891)				
	Examination for Defects	20		3.4	4.4.2.4
	Stud, Spring (Dwg. 8883721)				
	Examination for Defects	10		3.4	4.4.2.5

NOTES:

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TABLE II. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-F-48877(AR)

PARAGRAPH	TITLE: Fuze Rocket, M439		SHEET 2 OF 10		DRAWING NUMBER: Listed below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Rotor Housing (Dwg. 9260795) Examination for Defects	10		3.4	4.4.2.16
	Socket, Detonator (Dwg. 9260799) Examination for Defects	25		3.4	4.4.2.7
	Ground Contact, S&A (Dwg. 9352734) Examination for Defects	25		3.4	4.4.2.8
	Contact, Common (Dwg. 9260783) Examination for Defects	25		3.4	4.4.2.9
	Contact, N.C. (Dwg. 9260785) Examination for Defects	25		3.4	4.4.2.10

NOTES:

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PARAGRAPH	TITLE: Fuze Rocket, M439		SHEET 3 OF 10		DRAWING NUMBER: Listed below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Contact, Detonator (Dwg. 9260787) Examination for Defects	25		3.4	4.4.2.11
	Contact, Ground (Dwg. 9260788) Examination for Defects	25		3.4	4.4.2.12
	Contact, N.O. (Dwg. 9260784) Examination for Defects	25		3.4	4.4.2.13
	Plate, Top (Dwg. 9260781) Examination for Defects	25(a)		3.4	4.4.2.14
	Plate, Bottom (Dwg. 9260782) Examination for Defects	25(a)		3.4	4.4.2.15
	Gear and Pinion Assembly No. 1 (Dwg. 8883703) Examination for Defects	25		3.4	4.4.2.16
	Load Test of No. 1 Gear and Pinion Assembly	25		3.4	4.5.6.1

NOTES:

(a)Minimum of 5 per cavity for multiple mold with at least 25 total.

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TABLE II. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 4 OF 10		DRAWING NUMBER
	Fuze Rocket, M439				Listed below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Gear and Pinion Assembly No. 2 (Dwg. 8883700)				
	Examination for Defects	25		3.4	4.4.2.17
	Load Test of No. 2 Gear and Pinion Assembly	25		3.4	4.5.6.2
	Escapement Gear and Pinion Assembly (Dwg. 8883710)				
	Examination for Defects	25		3.4	4.4.2.18
	Load Test of Escapement Gear and Pinion Assembly	25		3.4	4.5.6.3
	Lever Assembly (Dwg. 9260776)				
	Examination for Defects	30		3.4	4.4.2.19
	Load Test of Lever Shaft	10		3.4	4.5.6.4
	Load Test of Lever Pallet	10		3.4	4.5.6.5
	Torque Test of Lever Shaft	10		3.4	4.5.6.6
	Rotor Assembly (Dwg. 9260796)				
	Examination for Defects	20		3.4	4.4.2.20
	Load Test of Annular Gear	10		3.4	4.5.6.7
	Load Test of Lock Roller Pin	10		3.4	4.5.6.8

NOTES:

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TABLE II. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 5 OF 10	DRAWING NUMBER
	Fuze Rocket, M439		Listed below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Rotor Housing Assembly (Dwg. 9260794)				
	Examination for Defects	10		3.4	4.4.2.21
	Load Test of Spring Stud	10		3.4	4.5.6.9
	Inner Plate Assembly (Dwg. 9260804)				
	Examination for Defects	20		3.4	4.4.2.22
	Load Test of Bridge Rivet	20		3.4	4.5.6.10
	Escapement Assembly (Dwg. 9260774)				
	Examination for Defects	20		3.4	4.4.2.23
	Load Test of Pillars	20		3.4	4.5.6.11

NOTES:

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TABLE II. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 6 OF 10		DRAWING NUMBER
	Fuze Rocket, M439				Listed below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Setback Weight Assembly (Dwg. 9260790)				
	Examination for Defects	25		3.4	4.4.2.24
	Pull Off Test of Insulator	25		3.4	4.5.7
	Pull Off Test of Switch Pin	25		3.4	4.5.8
	Bottom Plate Assembly (Dwg. 9260780)				
	Examination for Defects	25		3.4	4.4.2.25

NOTES:

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PARAGRAPH	TITLE: Fuze Rocket, M439	SHEET 7 OF 10		DRAWING NUMBER: Listed below
				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	S&A Assembly w/o Switch (Dwg. 9260779)				
	Examination for Defects	25		3.4	4.4.2.26
	Setback Weight Return Capability	25		3.7	4.5.9
	Rotor Reset Capability	25		3.8	4.5.10
	Torque Test of Rotor Stop Stud	25		3.4	4.5.11
	S&A Assembly (M439 Rocket Fuze) (Dwg. 9260778)				
	Examination for Defects	45		3.4	4.4.2.27
	Load Test of Detonator	45		3.14.1	4.5.12
	Continuity through Detonator	45		3.14.2	4.5.13
	Non-Arming (a)	45		3.14.3	4.5.14, 4.5.15
	S&A Arming Time (a)	45		3.14.4	4.5.14, 4.5.16
	S/A Housing (Dwg. 9260699)				
	Examination for Defects	25		3.4	4.4.2.28
	Pin, Interfacing (Dwg. 9260697)				
	Examination for Defects	10(b)		3.4	4.4.2.29

NOTES:
(a) This test may be performed with a simulated detonator.
(b) Quantity of 10 for each type required.

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TABLE II. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 8 OF 10		DRAWING NUMBER
	Fuze Rocket, M439				Listed below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Electronic Assembly (Dwg. 9260689)				
	Examination for Defects	25		3.4	4.4.2.30
	Module Turn On Time	25		3.15.1	4.5.17
	Module Time Delay	25		3.15.2	4.5.18(a)
	Molded Electronics Assembly (Dwg. 9260700)				
	Examination for Defects	25(b)		3.4/ 3.1.1	4.4.2.31
	Module Turn On Time	810		3.15.1	4.5.17/Table VI
	Module Time Delay	810		3.15.2	4.5.18/Table VI
	Thermal Shock	60		3.16	4.5.19/Table VI
	Umbilical Ground Pin (Dwg. 9260719)				
	Examination for Defects	25		3.4	4.4.2.32
	Umbilical Charge Pin (Dwg. 9260720)				
	Examination for Defects	25		3.4	4.4.2.33

NOTES:
(a) Perform at ambient only
(b) Minimum of 5 per cavity for multiple mold with at least 25 total

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PARAGRAPH	TITLE		SHEET 9 OF 10		DRAWING NUMBER
	Fuze Rocket, M439				~~Listed below~~ NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Electronics Housing (Dwg. 9260698)				
	Examination for Defects	25(a)		3.4/ 3.1.1	4.4.2.34
	Electronics Housing Assembly (Dwg. 9260721)				
	Examination for Defects	25		3.4	4.4.2.35
	Corrosion Resistance of Protective Coating	25		3.4	
	Umbilical Assembly Pull Test	25		3.4	

NOTES:

(a) Minimum of 5 per cavity for multiple mold with at least 25 total.

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TABLE II. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE: Fuze Rocket, M439		SHEET 10 OF 10		DRAWING NUMBER: Listed below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Fuze Rocket, M439 (Dwg. 9260704)				
	Examination for Defects	246		3.4	4.4.2.36
	S&A Position Check	246		3.18	4.5.20
	Charge circuit continuity	246		3.19	4.5.21
	Temperature and Humidity	20		3.20.1	4.5.22
	Thermal Shock	36		3.20.2	4.5.23
	Jolt	12		3.21.1	4.5.24
	Jumble	12		3.21.1	4.5.24
	Forty Foot Drop	15		3.21.2	4.5.25
	Five Foot Drop	15		3.21.3	4.5.26
	Non-Environmental Control	68		3.22	4.5.27
	Helicopter Vibration	80		3.22	4.5.28
	Charge Circuit Continuity (Following helicopter vibration)	80		3.19	4.5.21
	Non-Arm	32		3.21.4	4.5.30
	Arming Time (Under Minimum)	32		3.23	4.5.31
	Arming Time (Over Max. or Fail to Arm)	32		3.23	4.5.31
	Mechanical Interrupt Test	32		3.21.5	4.5.3.2
	Ballistic Firing	120		3.25/ 3.26	4.5.36

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of loaded fuzes shall contain:

a. Detonators from not more than three detonator lots with the same interfix number from one manufacturer.

b. Metal parts bearing the same metal parts lot interfix number from one metal parts manufacturer.

4.4.2 Examination. See MIL-A-48078 and Tables VI and VII herein.

a. Sampling Plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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Quality Conformance Inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8883713
4.4.2.1	Spring, Detent				NEXT HIGHER ASSEMBLY 9260774
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Spring embrittlement	10		3.5	4.5.3
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Spring, Setback Weight	1 OF 1	8883718
			NEXT HIGHER ASSEMBLY 9260774

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Spring embrittlement	10		3.5	4.5.3
102	Setback weight spring set	10		3.6	4.5.4
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Strap, Rotor Ground				9260798
					NEXT HIGHER ASSEMBLY
					9260796
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Width at detonator end		0.40%	3.4	Gage
102	Strap improperly bent		0.40%	3.4	Gage/Visual
103	Width of detonator groove		0.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Speed Clip				9260891
					NEXT HIGHER ASSEMBLY
					9260778
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length of clip		0.40%	3.4	Gage
102	Diameter of clip		0.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Stud, Spring				8883721
					NEXT HIGHER ASSEMBLY
					9260794
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of stud, min.		0.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9260795 NEXT HIGHER ASSEMBLY 9260794
4.4.2.6	Rotor Housing				
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of base		0.40%	3.4	Gage
102	Rotor housing post separation		0.40%	3.4	Gage
103	True positin of eyelet holes, 2 places		0.40%	3.4	Gage
104	Diameter of eyelet holes, 2 places		0.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Socket, Detonator				9260799
					NEXT HIGHER ASSEMBLY
					9260796
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Detonator pin socket diameter		0.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877 (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.8	Ground Contact, S&A	1	1	9352734
				NEXT HIGHER ASSEMBLY: 9260780

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Radius of Bend (70° -2°)		0.40%	3.4	Gage
102	Diameter of eyelet hole		0.40%	3.4	Gage
103	Thickness of strip		0.40%	3.4	Gage
104	Length of contact arm		0.40%	3.4	Gage
105	Poor quality of plating		0.40%	3.4	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9260783
4.4.2.9	Contact, Common				NEXT HIGHER ASSEMBLY 9260780
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Angle of bend ($83^{\circ} + 2^{\circ}$)		0.40%	3.4	Gage
102	Angle of bend ($63^{\circ} + 2^{\circ}$)		0.40%	3.4	Gage
103	True position of eyelet hole		0.40%	3.4	Gage
104	Diameter of eyelet hole		0.40%	3.4	Gage
105	Thickness of strip		0.40%	3.4	Gage
106	Poor quality of plating		0.40%	3.4	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9260785
4.4.2.10	Contact, N.C.				NEXT HIGHER ASSEMBLY 9260780
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of eyelet hole		0.40%	3.4	Gage
102	Angle of bend (90°)		0.40%	3.4	Gage
103	Thickness of strip		0.40%	3.4	Gage
104	Poor quality of plating		0.40%	3.4	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9260787
4.4.2.11	Contact, Detonator				NEXT HIGHER ASSEMBLY 9260780
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Angle of bend ($49^{\circ} + 2^{\circ}$)		0.40%	3.4	Gage
102	Angle of bend ($86^{\circ} + 3^{\circ}$)		0.40%	3.4	Gage
103	Diameter of eyelet hole		0.40%	3.4	Gage
104	Thickness of strip		0.40%	3.4	Gage
105	Poor quality of plating		0.40%	3.4	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9260788
4.4.2.12	Contact, Ground				NEXT HIGHER ASSEMBLY 9260780
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Angle of bend ($90^{\circ} + 2^{\circ}$)		0.40%	3.4	Gage
102	True positin of eyelet hole		0.40%	3.4	Gage
103	Diameter of eyelet hole		0.40%	3.4	Gage
104	Thickness of strip		0.40%	3.4	Gage
105	Poor quality of plating		0.40%	3.4	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Contact, N.O.				9260784
					NEXT HIGHER ASSEMBLY 9260780
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of eyelet hole		0.40%	3.4	Gage
102	Diameter of eyelet hole		0.40%	3.4	Gage
103	True positin of dimple locator hole		0.40%	3.4	Gage
104	Angle of bend (90°)		0.40%	3.4	Gage
105	Thickness of strip		0.40%	3.4	Gage
106	Poor quality of plating		0.40%	3.4	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Plate Top				9260781
					NEXT HIGHER ASSEMBLY
					9260778
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Mold inspection	5/cav	0.40%	3.4	4.5.2
102	True position of eyelet holes, 2 places		0.40%	3.4	Gage
103	Diameter of eyelet holes, 2 places		0.40%	3.4	Gage
104	True position of eyelet bosses, 2 places		0.40%	3.4	Gage
105	True position of S&A ground contact hole		0.40%	3.4	Gage
106	True position of contact locator holes, 4 places		0.40%	3.4	Gage
107	True position of dimple locator holes, 3 places		0.40%	3.4	Gage
108	Location of common contact post		0.40%	3.4	Gage
109	Depth of Detonator contact cavity		0.40%	3.4	Gage
110	Outside diameter		0.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship places		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877 (AR)

PARAGRAPH 4.4.2.15	TITLE Plate Bottom		SHEET 1 OF 1		DRAWING NUMBER 9260782
					NEXT HIGHER ASSEMBLY 9260780
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Mold inspection	5/cav		3.4	4.5.2
102	True position of eyelet holes, 2 places		0.40%	3.4	Gage
103	Diameter of eyelet holes, 2 places		0.40%	3.4	Gage
104	True position of contact eyelet holes, 5 places		0.40%	3.4	Gage
105	Diameter of contact eyelet holes, 5 places		0.40%	3.4	Gage
106	True position of contact dimples, 6 places		0.40%	3.4	Gage
107	Outside diameter		0.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.16	Gear and Pinion Assembly No. 1				8883703
					NEXT HIGHER ASSEMBLY 9260774
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Maximum gap between gear and pinion		100%	3.4	Gage
2	Load test of No. 1 gear and pinion assembly		100%	3.4	4.5.6.1
Major	None defined				
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8883700
4.4.2.17	Gear and Pinion Assembly No. 2				NEXT HIGHER ASSEMBLY 9260774
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Maximum gap between gear and pinion		100%	3.4	Gage
2	Load test of No. 2 gear and pinion assembly		100%	3.4	4.5.6.2
Major	None defined				
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8883710
4.4.2.18	Escapement Gear and Pinion Assembly				NEXT HIGHER ASSEMBLY 9260774
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Maximum gap between gear and pinion		100%	3.4	Gage
2	Load test of escapement gear and pinion assembly		100%	3.4	4.5.6.3
Major	None defined				
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.19	Lever Assembly				9260776
					NEXT HIGHER ASSEMBLY 9260774
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Maximum gap between lever and shaft		100%	3.4	Gage
2	Load test of lever shaft		100%	3.4	4.5.6.4
3	Load test of lever pallet		100%	3.4	4.5.6.5
Major					
101	Torque test of lever shaft		100%	3.4	4.5.6.6
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.20	Rotor Assembly				9260796
					NEXT HIGHER ASSEMBLY 9260779
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Load test of annular gear		100%	3.4	4.5.6.7
Major					
101	Load test of lock roller pin		0.40%	3.4	4.5.6.8
102	Banking stud pullout test		0.40%	3.4	Gage
103	Rotor ground strap and detonator socket contact resistance		0.40%	3.4	Milliohmmeter
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9260794
4.4.2.21	Rotor Housing Assembly				NEXT HIGHER ASSEMBLY 9260779
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Load test of spring stud		0.40%	3.4	4.5.6.9
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9260804
4.4.2.22	Inner Plate Assembly				NEXT HIGHER ASSEMBLY 9260774
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Load test of bridge rivet		0.40%	3.4	4.5.6.10
102	Pillar push test		0.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.23	Escapement Assembly				9260774
					NEXT HIGHER ASSEMBLY 9260779
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Load test of pillars		0.40%	3.4	4.5.6.11
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.24	Setback Weight Assembly				9260790
					NEXT HIGHER ASSEMBLY 9260779
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Switch pin insulator - missing, scored, with voids or cracks		0.40%	3.4	Visual
102	Perpendicularity of Switch Pin		0.40%	3.4	Gage
103	Overall height of insulator from side of weight		0.40%	3.4	Gage
104	Shrinkage requirement		0.40%	3.4	Visual
105	Pull-off test of insulator	(a)	0.40%	3.4	4.5.7
106	Pull-off test of Switch Pin	(a)	0.40%	3.4	4.5.8
Minor					
201	Evidence of poor workmanship		0.65%	3.27	Visual

NOTES:

(a) The Pull-off test of the insulator shall be performed before the Pull-off test of the Switch Pin. The same sample units shall be used for each test.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9260780
4.4.2.25	Bottom Plate Assembly				NEXT HIGHER ASSEMBLY 9260778
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Eyelets above flush, 5 places		0.40%	3.4	Visual
102	Alignment of N.C. contact		0.40%	3.4	Visual
103	Alignment of common contact		0.40%	3.4	Visual
104	Alignment of N.O. contact		0.40%	3.4	Visual
105	Alignment of ground contacts		0.40%	3.4	Gage
106	Loose connectors		0.40%	3.4	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.26	S&A Assembly w/o Switch		9260779
			NEXT HIGHER ASSEMBLY 9260778

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Distance between annular gear and No. 2, gear, max.		100%	3.10	Gage
2	Setback weight return capability		100%	3.7	4.5.9
3	Rotor reset capability		100%	3.8	4.5.10
Major					
101	Height from upper surface of base to top of rotor, max.		0.40%	3.9	Gage
102	Detonator socket contact missing		0.40%	3.4	Visual
103	Rotor ground strap missing		0.40%	3.4	Visual
104	Rotor ground strap deformed		0.40%	3.4	Visual
105	Rotor ground strap not making contact with detonator socket contact		100%	3.4	Visual
106	Foreign matter present		0.40%	3.4	Visual
107	Lubricant missing		0.40%	3.4	Visual
108	Diameter across outside corners of posts of rotor housing min.		0.40%	3.4	Gage
109	Protective finish missing or incomplete		0.40%	3.4	Visual

NOTES:

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MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9260779
4.4.2.26	S&A Assembly w/o Switch Continuation Sheet				NEXT HIGHER ASSEMBLY 9260778
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
110	Markings missing or incomplete		0.40%	3.4	Visual
111	Rotor Assembly does not lock in fully armed position.		0.40%	3.4	Visual
112	Alignment of detonator cavity with the outside diameter		0.40%	3.11	Gage
113	Torque test of rotor stop stud		0.40%	3.4	4.5.11
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER 9260778
4.4.2.27	S&A Assembly (M439 Rocket Fuze)				NEXT HIGHER ASSEMBLY 9260704
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Non arming (a)		100%	14.3	4.5.15
Major					
101	Setback weight actuator pin improperly positioned on common contact		0.40%	3.4	Visual
102	Overall length, max./min.		0.40%	3.4	Gage
103	S&A improperly assembled and seated in top plate		0.40%	3.4	Visual
104	Contact resistance between #1 & #5 contacts (safe)		100%	3.12	Milliohmmeter
105	Insulation resistance of #1 & #2 contacts (safe)		100%	3.12	Megohmmeter, test voltage 500 VDC

NOTES:

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MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 9260778
4.4.2.27	S&A Assembly (M439 Rocket Fuze)				NEXT HIGHER ASSEMBLY 9260704
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major -	Cont'd				
106	Insulation resistance of #1 & #5 contacts (armed w/o detonator)		100%	3.12	Megohmmeter, test voltage 500 VDC
107	Contact resistance between #1 & #2 contacts (armed w/o detonator)		100%	3.12	Milliohmmeter
108	Contact resistance between #4 & #6 contacts & rotor (armed w/o detonator)		100%	3.12	Milliohmmeter
109	Speed clip protrudes beyond outer periphery of rotor after assembly		0.40%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.25	S&A Assembly (M439 Rocket Fuze)	3 OF 3	9260778
			NEXT HIGHER ASSEMBLY 9260704

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major - Cont'd					
110	Detonator shorting switch not contacting detonator high terminal		0.40%	3.4	Visual
111	Distance from detonator disc to outside diameter of S&A rotor, max. (armed position)		0.40%	3.4	Gage
112	Detonator perforated or damaged (c)		0.40%	3.4	Visual
113	Load test of Detonator		0.40%	3.14.1	4.5.12
114	Continuity through detonator (b)		100%	3.14.2	4.5.13
115	Arming Time (e)		100%	3.14.4	4.5.16 (d)
Minor					
201	Marking missing or incomplete		0.65%	3.4	Visual
202	Evidence of poor workmanship		0.65%	3.28	Visual

(a) This shall be last test prior to insertion of the S&A Assy into the Fuze Dwg. 9260704.
(b) This test shall consist of arming S&A, measuring continuity, returning rotor to safe position.
(c) Detonators are to be tested in accordance with Para. 4.5.5.
(d) Perform at ambient temperature only.
(e) May be performed with a simulated detonator.

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9260699
4.4.2.28	S/A Housing				NEXT HIGHER ASSEMBLY 9260704
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Mold Inspection	5/ cavity		3.4	4.5.2
102	Length from forward end of housing to aft step		0.40%	3.4	Gage
103	Rotor outside diameter clearance radius		0.40%	3.4	Gage
104	Inside diameter of housing		0.40%	3.4	Gage
105	Outside diameter of housing flange		0.40%	3.4	Gage
106	Depth of S&A cavity		0.40%	3.4	Gage
107	Forward inside radius (2 places)		0.40%	3.4	Gage
108	Width between rotor housing guides		0.40%	3.4	Gage
109	Width of index slots		0.40%	3.4	Gage
110	Thickness of index slots		0.40%	3.4	Gage
111	Outside diameter of housing		0.40%	3.4	Gage
112	Width between rotor housing guides		0.40%	3.4	Gage
113	Thickness of rotor housing guides		0.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship/molding		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH 4.4.2.29	TITLE Pin Interfacing (Applicable Pins only)		SHEET 1 OF 1		DRAWING NUMBER 9260697
					NEXT HIGHER ASSEMBLY 9260689
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Longer length after bending		0.40%	3.4	Gage
102	Shorter length after bending		0.40%	3.4	Gage
103	Diameter of wire		0.40%	3.4	Gage
104	Wire tip flat over max.		0.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.30	Electronic Assembly				9260689
					NEXT HIGHER ASSEMBLY 9260700
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Defective soldered connections		0.40%	3.4	Visual
102	Components improperly mounted on substrate		0.40%	3.4	Visual
103	Components missing from substrate		0.40%	3.4	Visual
104	Coating missing, incomplete or defective		0.40%	3.4	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.31	Molded Electronics Assembly		9260700
			NEXT HIGHER ASSEMBLY 9260704

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Mold inspection	5/ cavity		3.4	4.5.2
102	Length of interface pins from mold at center line of pins		0.40%	3.4	Gage
103	True position of pins to reference diameter A		0.40%	3.4	Gage
104	Outside diameter of mold, min.		0.40%	3.4	Gage
105	Chamfer at forward banking surface		0.40%	3.4	Gage/Visual
106	Overall length		0.40%	3.4	Gage
107	Height of soldered pin from contact forward surface, max.		0.40%	3.4	Gage
108	Module turn on time		0.40%	3.15.1	4.5.17
109	Function time delay: 2 seconds		100%	3.15.2	4.5.18(a)
110	Function time delay: 8 seconds		100%	3.15.2	4.5.18(a)
111	Function time delay: 12 seconds		100%	3.15.2	4.5.18(a)
112	Function time delay: 16 seconds		100%	3.15.2	4.5.18(a)
113	Function time delay: 24 seconds		100%	3.15.2	4.5.18(a)
114	Thermal Shock	30		3.16	4.5.19(a)

NOTES:
(a)Perform at ambient only

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9260700
4.4.2.31	Molded Electronics Assembly (Continued)				NEXT HIGHER ASSEMBLY 9260704
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Pins not clean of mold material or have improper spherical termination		0.65%	3.4	Visual
202	Mold cavity identification not permanent or illegible		0.65%	3.4	Visual
203	Length of capacitor shorting pins		0.65%	3.4	Gage
204	Molding defects		0.65%	3.28	Visual
205	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9260719
4.4.2.32	Umbilical Ground Pin				NEXT HIGHER ASSEMBLY 9260721
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of interface connector		0.40%	3.4	Gage
102	Outside diameter of umbilical connector pin		0.40%	3.4	Gage
103	Termination radius of switch assy. connector pin		0.40%	3.4	Visual
104	True position of umbilical connector pin to interface connector pin		0.40%	3.4	Gage
Minor					
201	Absence of pin chamfer		0.65%	3.4	Visual
202	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9260720
4.4.2.33	Umbilical Charge Pin				NEXT HIGHER ASSEMBLY 9260721
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of interface connector pin		0.40%	3.4	Gage
102	Outside diameter of umbilical connector pin		0.40%	3.4	Gage
103	Radius termination of interface		0.40%	3.4	Gage
104	True postion of umbilical connector pin to interface connector pin		0.40%	3.4	4.5.2
Minor					
201	Width of seal slot		0.65%	3.4	Gage
202	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.34	Electronics Housing				9260698
					NEXT HIGHER ASSEMBLY 9260721
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Mold inspection	5/ cavity		3.4	4.5.2
102	True position of charge pin hole diameter to diameter A		0.40%	3.4	Gage
103	True position of ground pin hole diameter to diameter A		0.40%	3.4	Gage
104	True position of seal slot inside diameter to diameter A		0.40%	3.4	Gage
105	Inside diameter of cavity		0.40%	3.4	Gage
106	Diameter of charge pin hole		0.40%	3.4	Gage
107	Diameter of ground pin hole		0.40%	3.4	Gage
Minor					
201	Width of seal slot		0.65%	3.4	Gage
202	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.35	Electronic Housing Assembly				9260721
					NEXT HIGHER ASSEMBLY 9260704
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Protrusion of connectors above flange		0.40%	3.4	Gage
102	True position of charge pin to outside diameter of housing		0.40%	3.4	Gage
103	True position of ground pin to outside diameter of housing		0.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.36	Fuze, Rocket, M439	1 OF 2	9260704
			NEXT HIGHER ASSEMBLY: -------

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	S&A position check		100%	3.17	4.5.20
2	Jolt	12		3.21.1	4.5.24
3	Jumble	12		3.21.1	4.5.24
4	Forty-Foot Drop	15(b)		3.21.2	4.5.25
5	Five Foot Drop	15(c)		3.21.3	4.5.26
6	Non-environmental control	36		3.22	4.5.27
7	Helicopter Vibration	65		3.22	4.5.28
8	Non-arm	32(a)		3.21.4	4.5.30
9	Arming time (under minimum)	32(a)		3.23	4.5.31
10	Mechanical interrupt	32(a)		3.21.5	4.5.32
Major					
101	Length of fuze, max.		0.40%	3.4	Gage
102	Diameter of fuze, max.		0.40%	3.4	Gage
103	Charge circuit continuity		100%	3.19	4.5.21
104	Arming time (over max. or fail to arm)	32(a)		3.23	4.5.31
105	Temperature and Humidity	20(b)		3.20.1	4.5.22
106	Charge circuit continuity (following helicopter vibration)	65		3.19	4.5.21

NOTES:
(a) Samples taken from Helicopter vibration & Non-environmental.
(b) First article only.
(c) Five foot drop not done on every lot.

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MIL-F-48877(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.36	Fuze, Rocket, M439				9260704
					NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major - Continued					
107	Thermal Shock	36(b)		3.20.2	4.5.23
108	Ballistic firing	64(a)		3.26	4.5.36
Minor					
201	Stamping missing		0.65%	3.4	Visual
202	Evidence of poor workmanship		0.65%	3.28	Visual

NOTES:
(a) Samples taken from Helicopter vibration & Non-environmental.
(b) First article only.
(c) Five foot drop not done on every lot.

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4.4.3 Testing. Testing is described in the quality conformance inspection tables starting with 4.4.2.1.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6.2 of MIL-A-48078 and section 6.5 herein. The contractor shall submit all inspection equipment designs to the government for evaluation and approval prior to fabrication of such inspection equipment.

4.4.4.1 Contractor inspection system. As part of contractor's approved quality procedure, there shall be a provision for checking the adequacy of inspections made in connection with listed critical or major defects. These procedures shall include the following provisions:

4.4.4.1.1 Handling procedure. A Design Agency approved procedure shall be instituted that shall preclude unauthorized handling of the S&A's and fuzes from the S&A position check until the S&A has been installed in the fuze. The procedure shall be fail safe to eliminate armed fuzes from being introduced into the lot. Fail safe is defined as follows: Any equipment failure shall operate to cause rejection of the part or otherwise prevent its subsequent use.

4.4.4.1.2 Identification of rejected material. All material listed below, rejected as containing defects, shall be isolated to preclude its inclusion with acceptable material:

a. All S&A's failing paragraphs 4.5.9, 4.5.10, 4.5.15, 4.5.16, or 4.5.20.

b. All modules failing paragraphs 4.5.17, 4.5.18, or 4.5.19.

c. All fuzes failing paragraphs 4.5.24, 4.5.25, 4.5.26, 4.5.28, or 4.5.29.

4.4.4.2 Use of simulated defect standards. The following listed verified inspections shall be validated and inspection equipment verified once per shift using Design Agency approved item facsimilies or standards containing the defects and supplied by the contractor. The contractor shall resupply the item facsimilies when it is determined by the Government that the facsimilies are no longer representative enough of the product to

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make it unidentifiable to the contractor inspection personnel, or it is not functioning in the necessary manner and cannot be corrected satisfactorily. These standards should be inert and readily identifiable to the Government to ensure recovery should they be inadvertently introduced into production. Simulated defect standards used to verify the following verified inspections should be constructed and identified in such a manner that they are not readily recognizable to the contractor inspection personnel as defect standards. Standards should be used so that their number, time and sequence of introduction into the inspection cycle is known only to those personnel responsible for checking the accuracy of the operation. Adequacy of the standards shall be verified at least once each week.

Verified Inspection

Paragraph 4.5.9 - Setback Weight Return Capability.
Paragraph 4.5.20 - S&A Position Check.
Paragraph 4.5.17 - Module Turn on Time.
Paragraph 4.5.18 - Module Time Delay Function Test.

NOTE: When there is automatic detection and automatic rejection of defective items to a controlled area or a controlled rejection of defective items, it is permissible for the defect standards to be identifiable and their introduction into production noticable to contractor inspection personnel.

4.4.4.3 Undetected defect standards. Provisions shall be established for the corrective action to be taken where the equipment or contractor inspection personnel fail to identify a standard as defective. These provisions shall include a reinspection of all acceptable material subsequent to the last acceptable verification check as a minimum.

4.5 Methods of inspection.

PRECAUTION: This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and

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found to comply with their respective drawings and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspection. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table II.

4.5.2 Dimensional control of molded or plastic parts. In place of normal sampling and inspection associated with the Classification of Defects, and after a curing time for the parts has been approved (see Note 1) a sample of five (5) parts as molded from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the sample, one (1) part shall be the last produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each type of defect according to MIL-STD-105, using the AQL's detailed in the classification of defects. All cavities removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the contracting officer to substantiate the request.

NOTE 1: In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to Commander, ARRADCOM Dover, NJ 07801, ATTN: DRDAR-QAF-T. If there is a change in material, or in the cycle time, or if a cavity is reworked, or a new cavity is used, a new curing time shall be established and approved.

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4.5.3 Spring embrittlement. The spring shall be stretched along its longitudinal axis in the case of a coil spring, or bent in the case of a flat spring until the spring no longer returns to its original position. When the spring no longer returns to its original length or profile, it shall be considered distorted permanently. Any spring that breaks prior to this point shall be classed defective. One defective shall be cause for lot rejection.

4.5.4 Setback weight spring set. The spring shall be measured at its original free length, assembled in a restraining fixture in accordance with 4.4.4 and compressed to a normal height of .750 inches, placed in an oven and subjected to a temperature of not more than 180 degrees Fahrenheit (^{O}F) or less than 175°F for 24 hours. The spring shall be removed from the oven and cooled to ambient temperature. After cooling the spring for 24 hours minimum, the spring shall be measured and the measurement shall be compared to the original free length to ascertain compliance with the specified requirement. Any spring that fails the requirement shall be classed defective. One defective shall be cause for lot rejection.

4.5.5 Check test for deterioration of detonators. If the total time between original acceptance of any detonator lot and the assembly of that lot into the fuzes exceeds two years, at any time since previous tests, the detonator lot shall be subjected to and must satisfactorily pass the functioning and output tests specified in MIL-D-45495 immediately before the detonator lot is assembled into the fuzes. The test shall be performed by the contractor on detonators randomly selected by a Government Inspector. The test shall be conducted at Government expense without cost to the contractor who loaded the detonators or the contractor who is to assemble the detonators into the fuzes. The test shall not constitute a basis for fuze lot rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc. permitted while the detonator lot was under the jurisdiction of either contractor.

4.5.6 S&A mechanical tests.

4.5.6.1 Load test of No. 1 gear and pinion assembly. Apply an axial load perpendicular to the No. 1 gear and in such a direction as to test the assembly of the No. 1 gear and pinion. The required force shall act against the No. 1 gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any No. 1 gear and pinion assembly that fails the requirement shall be classed defective and removed from the lot.

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4.5.6.2 Load test of No. 2 gear and pinion assembly. Apply an axial load perpendicular to the No. 2 gear and in such a direction as to test the assembly of the No. 2 gear and pinion. The required force shall act against the No. 2 gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any No. 2 gear and pinion assembly that fails the requirement shall be classed defective and removed from the lot.

4.5.6.3 Load test of escapement gear and pinion assembly. Apply an axial load perpendicular to the escapement gear and in such a direction as to test the assembly of the escapement gear and pinion. The required force shall act against the escapement gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any escapement gear and pinion assembly that fails the requirement shall be classed defective and removed from the lot.

4.5.6.4 Load test of lever shaft. Apply an axial load perpendicular to the lever and in such a direction as to test the assembly of the lever and shaft. The required force shall act against the lever. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any lever shaft that fails the requirement shall be classed defective and removed from the lot.

4.5.6.5 Load test of lever pallet. Apply an axial load perpendicular to the lever and in such a direction as to test the assembly of the lever and the lever pallet. The required force shall act against the lever. Upon removal of the required force, the height of the lever pallet above the lever shall be inspected for compliance with the drawing requirement. Any lever pallet that fails the requirement shall be classed defective and removed from the lot.

4.5.6.6 Torque test of lever shaft. A torque shall be applied between the lever and the lever shaft. Any lever shaft that does not comply with the drawing requirement shall be classed defective.

4.5.6.7 Load test of annular gear. Apply a load perpendicular to the rotor and in such a direction as to test the assembly of the annular gear studs to the annular gear and the rotor. The required force shall act against the rotor. Upon removal of the required force, the distance from the face of the rotor to the outside face of the annular gear shall be inspected for compliance with the drawing requirement. Any annular gear that fails the requirement shall be classed defective and removed from the lot.

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4.5.6.8 Load test of lock roller pin. Apply an axial load perpendicular to the rotor and in such a direction as to test the strength of the assembly of the lock roller pin to the rotor. The required force shall act against the rotor. Any lock roller pin that does not comply with the drawing requirement shall be classed defective.

4.5.6.9 Load test of spring stud. Apply an axial load perpendicular to the rotor housing and in such a direction as to test the assembly of the spring stud and the rotor housing. The required force shall act against the rotor housing. Any spring stud that does not comply with the drawing requirement shall be classed defective.

4.5.6.10 Load test of bridge rivet. Apply the required load perpendicular to the inner plate and into the bridge rivet from the side opposite the bridge. The required force shall act against the inner plate. Any bridge rivet that does not comply with the drawing requirement shall be classed defective.

4.5.6.11 Load test of pillars. Apply the required axial load perpendicular to the inner plate and in such a direction as to test the assembly of the pillars to the inner plate and outer plate. The required force shall act against the inner plate. Any pillars that do not comply with the drawing requirement shall be classed defective.

4.5.7 Pull-off test of insulator. Secure the setback weight in a holding fixture. An axial pull of a minimum of two pounds shall be applied to the pin assembly. Any insulator that separates from the switch pin shall be classed defective.

4.5.8 Pull-off test of switch pin. Secure the setback weight in a holding fixture. An axial pull of a minimum of five pounds shall be applied to the pin assembly. Any pin assembly that separates from the setback weight shall be classed defective.

4.5.9 Setback weight return capability. The S&A in the safe position shall be assembled to the test equipment in accordance with 4.4.4. The S&A shall be securely held in place during a 20 pound load test with a hold down force. The 20 pound load shall be applied at an average rate of 300 $\pm$ 50 inches per minute to the

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setback weight. It is mandatory that the setback weight maintain contact with the 20 pound load applicator until fully bottomed. The 20 pound force shall then be removed. S&A's failing to completely return to the fully safe position shall be classed defective and removed from the lot. Defect standards for verification of the inspection process shall be in accordance with 4.4.4.2 and 4.4.4.3. Defect standards for use with automatic equipment shall be as follows:

a. Units with setback weights which bind when depressed and remain in the depressed condition.

b. Units with setback weights which start to return but do not completely return to the original locking position.

c. Units with setback weights which return from the depressed condition at a slower than normal rate causing rejection.

4.5.10 Rotor reset capability. The loaded S&A shall be assembled to the test fixture in accordance with 4.4.4. The rotor shall be rotated to a position at least 26°, but not more than 35° from the fully safe position and released. Observation shall be made for rotor return to the fully safe position. S&A's failing to comply with the specified requirement shall be classed defective and removed from the lot.

4.5.11 Torque test of rotor stop stud (eccentric pin). The required torque shall be applied between the rotor stop stud and gear assembly. The required force shall act on the gear assembly. Any rotor stop stud that does not comply with the drawing requirement shall be classed defective.

4.5.12 Load test of detonator. Apply an axial load of 5 pounds perpendicular to the rotor and in such a direction as to test the strength of the speed clip holding the detonator in the rotor. Any speed clip that moves more than .005 inch shall be classed defective and the S&A removed from the lot.

WARNING: This test may be dangerous and should be performed behind a suitable barricade.

4.5.13 Continuity through detonator test. The S&A shall be armed in the case and the resistance measured between terminals three and four. The value shall be 3.5 ± 1.5 ohms. After the test the acceptable S&A's shall be disarmed and returned to the line for further assembly. Any S&A failing the requirement shall

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be classed defective and removed from the lot. Caution - Any continuity measurement which passes electrical current through the detonator shall be limited to one milliampere maximum.

4.5.14 Temperature conditioning. Prior to the centrifuge test (4.5.15 and 4.5.16), the sample shall be conditioned for a minimum of two hours after temperature stabilization at the following temperatures:

$70^{o} \pm 15^{o}F$ - 15 units
$-50^{o} \pm 5^{o}F$ - 15 units
$+160^{o} \pm 5^{o}F$ - 15 units

4.5.1.5 M439 S&A non-arming test. The S&A in the safe position shall be assembled to the test equipment in accordance with 4.4.4. The equipment shall accelerate the loaded Safety and Arming Device in a clockwise direction to 12.0 g's minimum for a minimum of five (5) seconds. Observation shall be made for rotor in the safe position. S&A's failing to comply with the specified requirement shall be classed defective and removed from the lot.

4.5.16 M439 S&A arming time test. The S&A in the safe position shall be assembled to the test equipment in accordance with 4.4.4. The equipment shall accelerate the Safety and Arming Device to 55g's ± 0.5g's in a direction which simulates rocket launch. The arming time shall be recorded to the nearest hundredth of a second using a calibrated timer. At the completion of acceleration, the equipment shall be halted and visual observation shall be made for the locking position of the detent; to ascertain positive arming. At the completion of the test, the Safety and Arming Device shall be returned to the safe position. Any S&A that fails the requirement shall be classed defective and removed from the lot.

WARNING: This test may be dangerous, and should be performed behind a suitable barricade.

4.5.17 Module turn on time. (See Figure 2). The module shall be tested by applying -16 VDC to pin 1 for 100 msec, waiting 7 seconds and then shorting pin 2 to pin 4. Any module that does not deliver an output pulse at pin 3 within 600 microseconds shall be classed defective.

4.5.18 Module time delay function test. (See Figure 2). A compensated voltage setter that represents the helicopter setter system shall be used. After shorting pins 5, 6 and 4 together for four seconds, the module shall be capable of accepting a setting charge from the compensated voltage setter and exhibiting a corresponding time delay as listed in Table VI. The time delay shall be the time interval from the time pin 2 is connected to pin

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4 and the leading edge of the output pulse exits pin 3. The firing output pulse (pin 3) shall provide at least 1000 ergs of energy in less than 100 microseconds into an M84 detonator or a resistor of 3.5 + 1.5 ohms. Record the time delay setting, function time delay and energy output (equal to, greater than or less than) by serial number or module lot no. Any module that does not meet the requirement shall be classed defective and removed from the lot.

4.5.19 Module thermal shock. The module shall be subjected to the Thermal Shock Test in accordance with MIL-STD-331, Test 113. These modules shall then be subjected to the timing test as specified in 4.5.18. Any module that does not meet the test requirements shall be classed defective. One defect shall be cause for lot rejection.

4.5.20 S&A position check. The S&A position, before insertion into the fuze, shall be inspected. Any armed S&A shall be considered defective and removed from the lot, and the assembly operation shall be halted until operation is corrected to Government satisfaction. The inspection shall be performed using procedures in accordance with 4.4.4.2 and 4.4.4.3 and equipment in accordance with 4.4.4.

Defect standards with semi-armed and fully armed conditions shall be introduced at various times to verify test performance. The Design Agency approved handling procedure to preclude untested S&A's from being introduced into the fuze lot shall also be assessed when the standards are introduced. If an improper procedure is being used as determined by the Government, all fuzes accepted subsequent to the last acceptable handling procedure assessment shall be screened as determined by the government.

4.5.21 Charge circuit continuity. The fuze shall be subjected to a charge circuit continuity test. Any fuze which is identified as a defect shall be removed from the lot or sample and held for failure analysis or rework. An open circuit continuity failure may indicate that the safety and arming device is armed; any fuze with this type of defect should be handled with appropriate safety precautions. The test shall be conducted using Government approved equipment and procedures.

4.5.22 Temperature and humidity. The fuze shall be subjected to the Temperature and Humidity Test per MIL-STD-331, Test 105.1 with the exception that only one 14 day cycle be used. Any fuze that does not meet MIL-STD-331 test requirements shall be classed defective.

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4.5.23 Thermal shock. The fuze shall be subjected to the Thermal Shock Test in accordance with MIL-STD-331, Test 113. These fuzes shall then be centrifuge fired. The fuze shall be charged to the nominal 24 second time before arming. After arming, the fuze shall be fired. Any fuze that does not meet test requirements shall be classed defective.

4.5.24 Jolt and jumble. The fuze shall be subjected to the Jolt Test specified in MIL-STD-331 with the exception that the fuze shall not be disassembled and inspected after the Jolt Test. After completion of the Jolt Test, the same fuze shall be subjected to and withstand the Jumble Test specified in MIL-STD-331. Any fuze that fails the requirements shall be cause for lot rejection.

4.5.25 Forty foot drop. The fuze with test warhead and motor shall be subjected to the Forty Foot Drop Test as specified in MIL-STD-331. The complete pack as specified on dwg. 9230116 and 9235840 shall be used. Any fuze that fails the requirement shall be classed defective.

4.5.26 Five foot drop test. A random sample of fifteen (15) fuzes shall be selected and subjected to the test as specified in MIL-STD-331 with the following conditions:

a. The fuze will be assembled to a test Warhead (the Government shall supply the warhead for the contractor's use).Three (3) fuzes must be dropped in each of the five orientations. No fuze shall be subjected to more than one drop. The orientations of the Safety and Arming Device (specifically the setback weight) of the fuze shall be random.

b. The Five Foot Drop Test, as specified in MIL-STD-331, shall be performed until three (3) consecutive lots (a minimum lot size of one week's production) have complied with the acceptance criteria specified. After satisfying the above requirement, one lot (a minimum lot size of one

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week's production) per month shall be selected by the Government and subjected to and pass the Five Foot Drop Test. Should the one lot (per month) fail the acceptance criteria, the contractor shall be required to return to the three (3) consecutive lot procedures above for process qualification.

c. The fuzes shall be subjected to the tests and acceptance or rejection criteria as specified in table III. Fuzes must be safe to handle and safe to fire or dispose of (not necessarily operable) following the Five Foot Drop Test and prior to performing the Non-arm and Arming Time Tests.

TABLE III. Acceptance criteria, five foot drop test.

Test	Accept	Reject	Test Method Paragraph
Non-Arm	0	1	4.5.30
Arming Time Under Min.	0	1	4.5.31

Any fuze that is unsafe to handle (See 6.6.1), fire or dispose of following the Five Foot Drop Test or does not meet the arming time under minimum definitions of 6.6.2 is classified as a safety failure. If the sample fails the above acceptance criteria or has one or more safety failures, the lot shall be rejected.

4.5.27 Non-environmental test. A random sample of twelve (12) fuzes shall be selected and subjected to the test and acceptance or rejection criteria as specified in 4.5.29.

4.5.28 Helicopter vibration test. Fuzes shall be subjected to Helicopter Vibration as specified in Figure 1. Vibration conditioning at resonance is not required. Following vibration conditioning, the fuzes shall be subjected to the tests and acceptance or rejection criteria specified in 4.5.29. NOTE: The fuzes shall be subjected to and must pass the continuity test (4.5.21) subsequent to the vibration test. If one or more continuity failures occur in the test sample the lot shall be rejected.

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4.5.29 Acceptance criteria. The sample fuzes from the Non-environmental (12) and Environmental tests (20) shall be combined into one sample and subjected to the tests specified in table IV and either Table V or Table VII, whichever is applicable. Arming time definitions are in 6.6.2.

TABLE IV. Acceptance criteria, non-environmental and vibration tests.

Test	Accept	Reject	Test Method Paragraph
Non-arm	0	1	4.5.30
Arming Time under min.	0	1	4.5.31
Arming Time over max.	0	1	4.5.31
Fail to arm	0	1	4.5.31
Mechanical interrupt	0	1	4.5.32

If all tests are acceptable, the sample fuzes from the Non-environmental test may be returned to the lot after disassembly, resetting the S&A into the safe position and reassembly.

4.5.30 M439 fuze non-arming test. The fuze shall be assembled to the test equipment in accordance with 4.4.4. The equipment shall accelerate the loaded fuze in a clockwise direction to 12.00g's plus or minus 0.5g's for a minimum of three (3) seconds. An examination shall be made for rotor in the safe position. Any fuze that fails the requirements of the non-arming test shall be classed defective.

4.5.31 M439 fuze arming time test. The fuze shall be assembled to the test equipment in accordance with 4.4.4. The equipment shall accelerate the loaded fuze in a clockwise direction to 55.0 g's plus or minus 0.5g's and the arming time shall be recorded to the nearest hundredth of a second using a swing fixture and acoustic pick-up feeding the arming cycle noise signal to a design agency approved recorder. Any fuze that fails the requirements of the arming time test shall be classed defective.

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NOTE: Removing the fuze from the cold temperature environment prior to performing the centrifuge test may cause frost to form on the metal parts. This may have an effect on the test results due to attenuation of the acoustic signal of the arming mechanism, thus making it difficult to distinguish the arm signal from background noise. The contractor shall take appropriate action to alleviate this potential problem. One effective method has been to seal each fuze in a plastic bag prior to temperature conditioning. Bagged fuzes are then subjected to the centrifuge test.

4.5.32 Mechanical interrupt test. The armed fuzes from 4.5.31 shall be checked for charge circuit continuity using Government approved equipment and procedures. Any fuze that does not indicate an open circuit between the charge circuit and the umbilical shall be classed defective.

4.5.33 Test equipment failure. If the test equipment fails to produce an arming readout (i.e., recording paper skewes, audio pick-up or amplifier inoperative) during the arming time test as verified by the Government, the Safety and Arming Device shall be visually inspected for arming; if found to be armed or partially armed, the Safety and Arming Device shall be reset into the unarmed position, reassembled into the same fuze body and retimed once as a fuze. The only handling action performed on the Safety and Arming Device shall be reset of the device just prior to the arming time test. The timing record produced by the fuze on the approved fuze timing equipment shall be considered the timing record of the fuze under the applicable defect classification.

4.5.34 Rounding of fuze arming times. Fuze arming times, if recorded to the thousandth of a second shall be "rounded off" to the nearest hundredth of a second.

4.5.35 Correlation factor. A correlation factor, which correlates the fuze arming time method (swing fixture) to the in-line S&A arming time method, shall be run for this fuze. Correlation tests shall be performed by the contractor with Government verification. Correlation factors shall be validated at six month intervals (maximum) and re-established whenever:

a. A new contract is issued
b. A change of fuze type
c. Fuze timing equipment is modified

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For the correlation procedure a digital counter or an approved equivalent device shall be used, set to read milliseconds for S&A timing and set for oscillograph traces, to the nearest millisecond, for the fuze. The correlation shall be accomplished as follows:

a. Select five (5) S&A's demonstrating acceptable arming time. Each S&A shall have repeatable arming times within ±25 milliseconds in five successive runs on the approved S&A timing equipment. S&A arming time readings shall be obtained from the first movement of the setback weight to the detent locking in the rotor.

b. The same S&A's shall be assembled to electronic modules and placed in fuzes. The fuzes shall then be timed five (5) times on the approved fuze timing equipment. Readings of the fuze arming time traces, from setback weight bottoming to rotor detent, shall be recorded.

c. The shift in average fuze time, if any, when compared with the average S&A time (taken as a standard) shall be used as the correlation correction and applied to the acceptance test results.

Data obtained in accordance with the above procedure shall be submitted to Commander, US Army Research and Development Command, Dover, NJ 07801, ATTN: DRDAR-QAF-T and DRDAR-QAF-I.

4.5.36 Ballistic firing test. The loaded M439 fuze shall be conditioned in accordance with table V or table VII, whichever is applicable. This test shall be performed at a Government facility using modified M229 test warheads. MK66 rocket motors shall be used for all tests and they shall be temperature preconditioned to 155°F ± 4°F prior to assembling to warhead/fuze. The fuze shall be set with a Government approved Ballistic Firing Test Set. A hard copy of the fuze charge signature shall be made at this time. Prior to setting and firing, the fuzes shall be checked for charge circuit continuity (4.5.21). Fuzes that fail the continuity test shall not be subjected to the ballistic test. Firing and fuze charging shall be accomplished by the Ballistic Firing Test Set using fuze time settings of 2, 8, 14 and 20 seconds. Time from launch to fuze function shall be detected and recorded.

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4.5.36.1 No-test conditions. Some ballistic firings may result in conditions that do not properly test fuze time delay function. If the following conditions occur during a test firing, that launch will be scored as a "no test."

a. The rocket hits the ground before the maximum time delay for that time setting occurs.

b. The fuze is not charged properly due to malfunctioning of the setter or the connecting cables.

4.5.36.2 Acceptance criteria.

Nominal Time Setting (Sec.)	Function Time Delay (Sec.)
2	1.87 to 2.13
8	7.84 to 8.16
14	13.72 to 14.28
20	19.60 to 20.40

4.5.36.3 Rejection criteria.

4.5.36.3.1 First article sample. If a total of three (3) or more units do not meet the function time delays of 4.5.36.2 or fail the charge circuit continuity test, the sample shall be rejected.

4.5.36.3.2 Lot acceptance sample. If a total of four (4) or more units do not meet the function time delays of 4.5.36.2 or fail the charge circuit continuity test, the lot shall be rejected.

4.5.37 Data items. The contractor shall submit a quality inspection test demonstration and evaluation report in accordance with Data Item Descriptions DI-R-1721 and DI-R-1724 giving the detailed test results.

5. PACKAGING

5.1 Packaging and marking. - Packaging and marking of fuzes shall be as specified on Dwg. 9270749.

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5.2 Packing and marking. - The packaged fuzes shall be packed and marked as specified on Dwg. 9270751 or 9270750, and shall comply with CFR49, Parts 100 - 199.

5.3 Ammunition data cards. Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Intended use. The fuzes covered by this specification are intended for the 2.75 Inch Rocket System.

6.2 Ordering data. See MIL-A-48078.

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.
4.5.37	Quality Inspection Test Demonstration Evaluation Report	DI-R-1724
4.5.37	Quality Inspection Defect Report	DI-R-1721

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.3.1 Defect reporting. From each lot produced, the contractor shall submit to Commander, ARRADCOM, ATTN: DRDAR-QAF-T, Dover, NJ 07801, results of the Module, Jolt, Jumble, Helicopter Vibration, and Centrifuge tests within three (3) days after completion of acceptance testing of the lot. If tests in addition to acceptance requirements are utilized to evaluate the lot, results of these tests will also be submitted three (3) days after the additional tests are completed. The defective fuzes shall be saved pending review of the results. See appendix for examples of test report formats.

6.4 First article. When a first article is required for inspection and approval, see 3.1 and 4.3.

6.5 Submission of inspection equipment designs for approval. See 6.2 of MIL-A-48078. Submit equipment designs as required to Commander, ARRADCOM, ATTN: DRDAR-QAF-I, Dover, NJ 07801.

6.6 Definitions.

6.6.1 Safe to handle. The fuze will be considered safe to handle if after electronic examination and tear down the following conditions are met:

a. The rotor shall be in the safe position and secured by the setback weight such that it cannot be turned to the armed position with finger pressure.

b. The lock roller pin on the rotor shall not be broken.

c. The setback weight shall be in the fully safe position (raised to its highest position).

d. No loose explosive shall be present in the fuze assembly.

e. No parts shall be broken or deformed which could present a safety hazard in handling or inadvertent firing.

6.6.2 Fuze arming and non-arming definitions.

a. Times under 0.70 second are classed as arming time under minimum.

b. Times 0.91 second through 1.20 seconds are classed as arming time over maximum.

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c. Times over 1.20 seconds or fuzes which do not arm or fuzes which time but do not detent are classed as failures to arm.

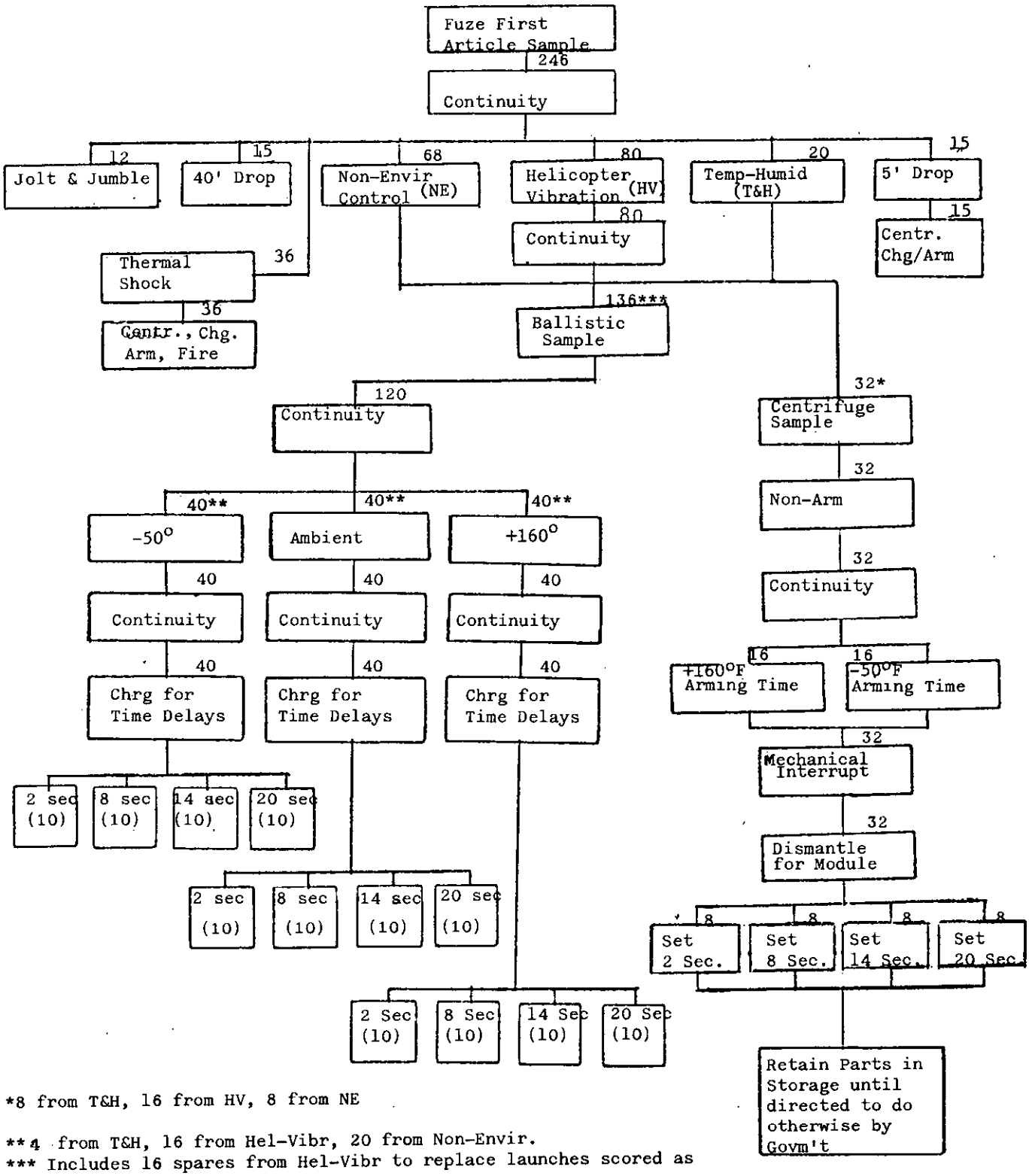
6.7 Government furnished parts.

6.7.1 Packaging and packing. The Procuring Agency will provide the packaging and the packing for the Forty Foot Drop.

6.7.2 Modified M229 test warhead. The Procuring Agency will provide Modified M229 test warheads for the forty foot drop, five foot drop and Ballistic Tests.

6.8 Source control and specification control drawings. Certificates of conformance from the supplier certifying that the item complies with the drawing requirements shall be provided to the Government representative.

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*8 from T&H, 16 from HV, 8 from NE

**4 from T&H, 16 from Hel-Vibr, 20 from Non-Envir.
*** Includes 16 spares from Hel-Vibr to replace launches scored as "No Tests".
NOTE: A setter that represents the helicopter setter system shall be used for delay time acceptance tests in all cases.

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TABLE VI. Module time delay test.

1. All modules shall be serialized.

2. A setter that represents the helicopter setter system shall be used for acceptance of all modules.

Nominal time settings	Environment	Module quantity	Acceptance limits (Sec)
2 Sec.	-65°F +70°F +160°F Thermal Shock	50 50 50 12	1.96 to 2.04
8 Sec.	-65°F +70°F +160°F Thermal Shock	50 50 50 12	7.84 to 8.16
14 Sec.	-65°F +70°F +160°F Thermal Shock	50 50 50 12	13.72 to 14.28
20 Sec.	-65°F +70°F +160°F Thermal Shock	50 50 50 12	19.60 to 20.40
24 Sec.	-65°F +70°F +160°F Thermal Shock	50 50 50 12	23.52 to 24.48

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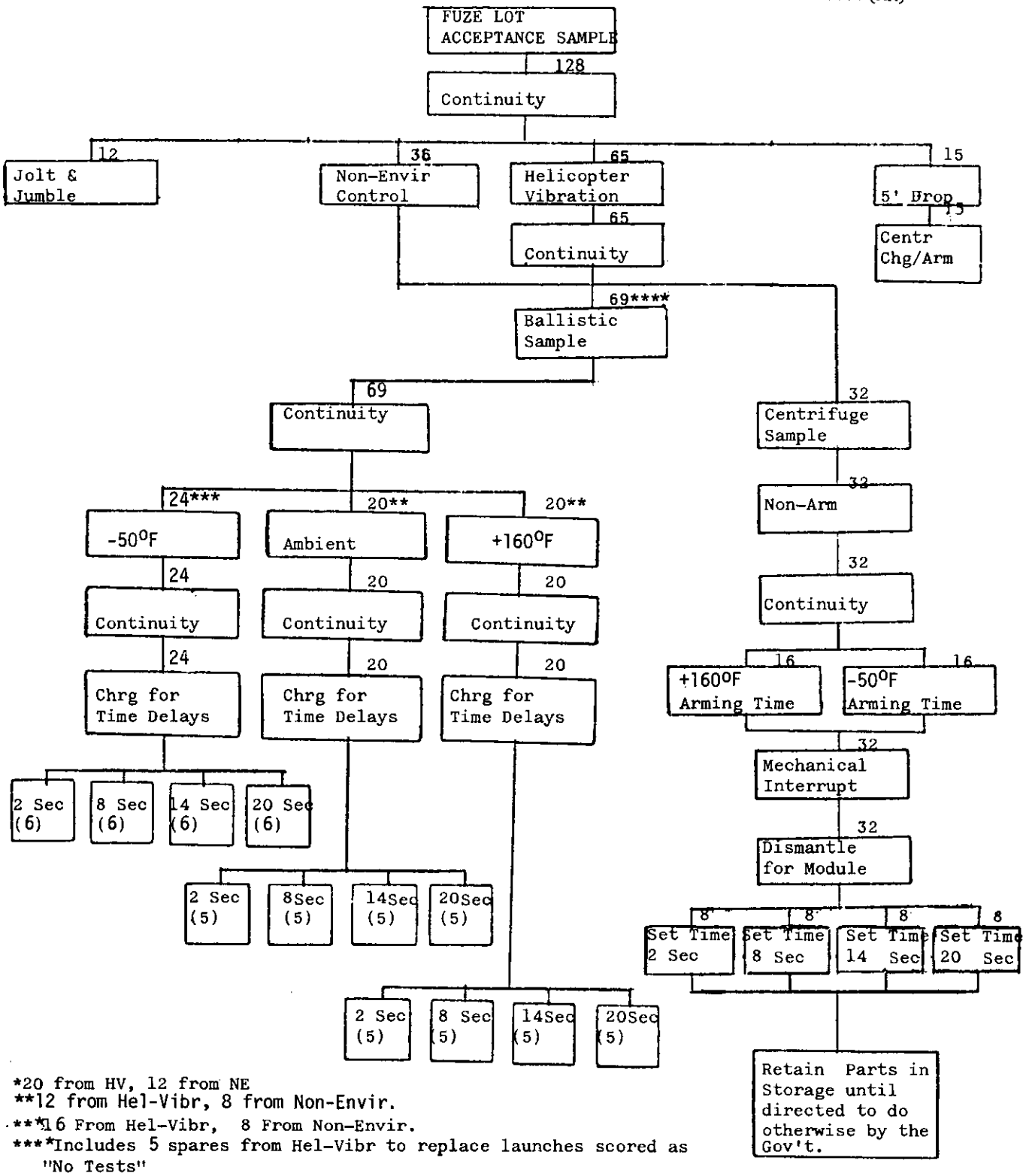
The firing capacitor shall be checked to insure that at least minimum detonator firing output pulse is present when the detonator firing circuit is tripped at the conclusion of the time delay.

3. All modules shall be checked for module turn on time at the temperatures specified in paragraph 2 above. Data obtained from these checks shall be recorded and be made traceable to individual modules.

4. The module timing circuit shall be tripped immediately following the charging of each module.

5. Modules shall be temperature conditioned for a minimum of two (2) hours after stabilization at the temperatures specified in paragraph 2 above.

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*20 from HV, 12 from NE
**12 from Hel-Vibr, 8 from Non-Envir.
***16 From Hel-Vibr, 8 From Non-Envir.
****Includes 5 spares from Hel-Vibr to replace launches scored as "No Tests"

NOTE: A setter that represents the helicopter setter system shall be used for delay time acceptance in all cases.

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One 15 minute sinusoidal sweep (5-500-5 HZ) shall be performed for each of three axes. No dwell time shall be spent on any one frequency. A search for resonance shall not be done.

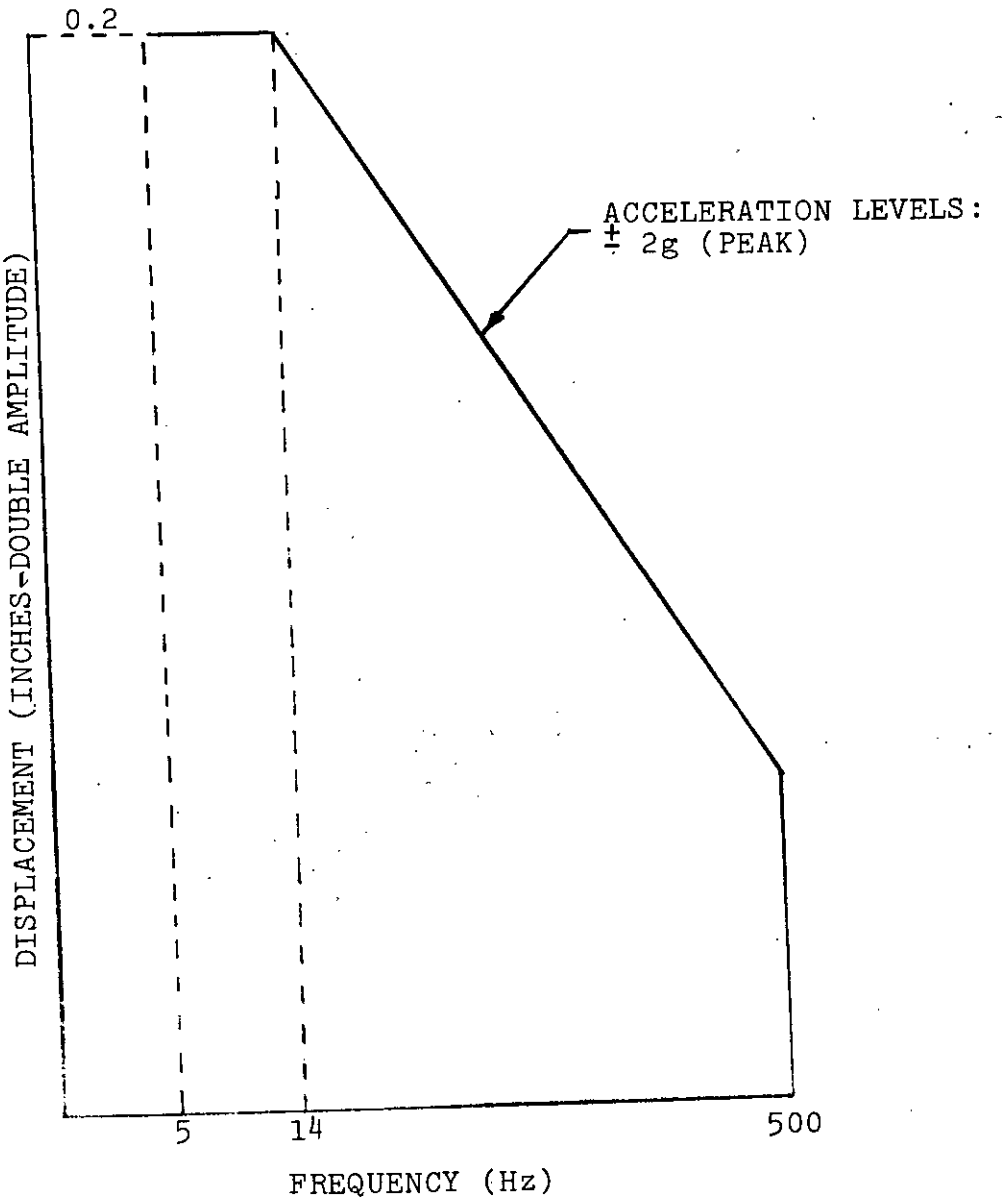
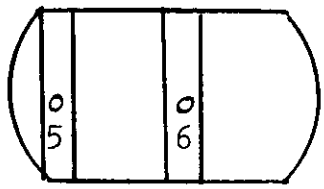
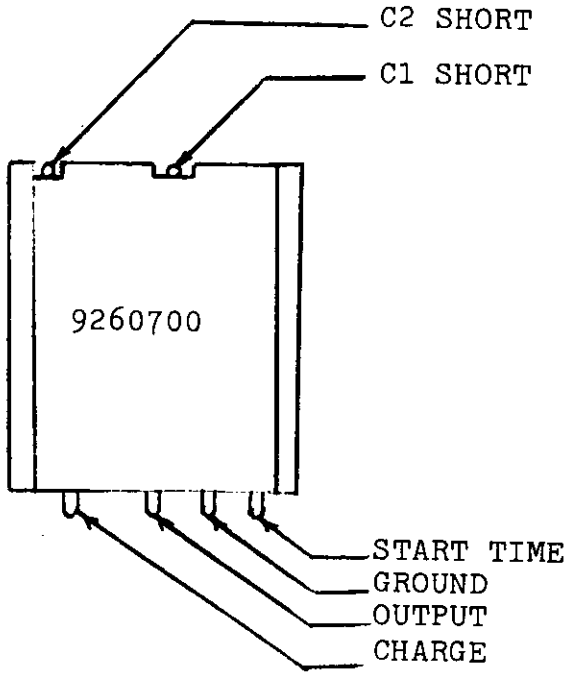


FIGURE 1. Helicopter vibration.

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CR1 = PIN #1 to C1-SHORT
CR2 = C2-SHORT TO PIN #1



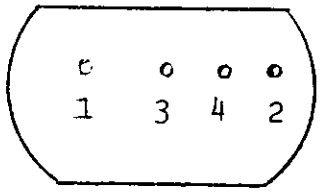


FIGURE 2. **Electronic module terminal diagram.**

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APPENDIX

TEST REPORT FORMATS

M_____METAL PARTS LOT ACCEPTANCE TEST DATA

Contractor _____ Fuze Lot No.: _____

Date of Test Lot Quantity (Gross): _____

Start: _____ Sample Size (Total) _____

End: _____ Sample Size

Returned to Lot: _____

Date Shipped: _____ Lot Quantity (Net): _____

Five Foot Drop Test

5' Drop

Sample Size	Serial No. of Defect	Failure Analysis

Non-Arm

Sample Size	Serial No. of Defect	Failure Analysis

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Arming Time

Sample	Number Accepted	Serial No. of Short Times	Serial No. of Long Times	Timed but Failed to Arm	Not Timed and failed to Arm	Armed but not Timed

Attach list of fuze serial numbers and arming times.

5' Drop Failure Analysis

Operational Tests

Vibration Test

Sample Size	Serial No. of Defects	Failure Analysis

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Fuze Non-Arm and Arming-Time Tests

Non-Arm Test

Sample Size	Serial No. of Defects	Failure Analysis

Arming Time Test

Sample Size	Number Accepted	Serial No. of Short Times	Serial No. of Long Times	Timed but Failed to Arm	Not Timed and failed Arm	Armed but not Timed

Attach list of fuze serial numbers and arming times.

Failure Analysis

Correlation applied to the arming time readings of the Five Foot Drop Test and the Operational Test samples was _____ Seconds.

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Module Time Delay Tests

Sample Size	Number Accepted	Serial No. of Short Times	Serial No. of Long Times	Serial No. of Failures to Function	Serial No. of Low Outputs

Jolt and Jumble

Sample Size	Serial No. of Defects	Failure Analysis

Failure Analysis

If lot failure is due to other than test conditioning.

_____
Government Inspector Date

_____
Contractor Representative Date

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Army-AR

Preparing activity:
Army-AR

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MIL-F-48877 Fuze, Rocket; M439 Parts and Assembly with Detonator

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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